T1] $\xi \sim R[0,\theta]$, $\theta \in \Theta$, выборка $\vec{x}_n$

a) 1) $\tilde{\theta}_1 = 2\bar{x} = 2\cdot\frac{1}{n}\sum_{i=1}^{n} x_{ni}$

2) $\tilde{\theta}_2 = x_{min}$, 3) $\tilde{\theta}_3 = x_{max}$, 4) $\tilde{\theta}_4 = x_1 + \frac{1}{n-1}\sum_{i=2}^{n} x_i$

$M\xi = \int_{-\infty}^{\infty} x\, dF(0,\theta) = \int_0^\theta x \frac{1}{\theta} dx = \frac{\theta}{2}$

$M\xi^2 = \int_0^\theta x^2 \frac{1}{\theta} dx = \frac{\theta^2}{3}$, $D\xi = M\xi^2 - (M\xi)^2 = \frac{\theta^2}{12}$

Для параметрической модели: $\forall\theta\in\Theta$ $M\tilde{\theta} = \theta$ – несмещенная оценка

$\forall\theta\in\Theta$ и $\forall\varepsilon>0$ $P(|\tilde{\theta}-\theta|\geq\varepsilon) \xrightarrow[n\to\infty]{} 0$ – состоятельная оценка

1) $\tilde{\theta}_1$: $\forall\theta>0$ $M\tilde{\theta}_1 = M(2\frac{1}{n}\sum_{i=1}^{n} x_i)$

x_i – незав. случайные величины с распредел. ξ

$\Rightarrow M\tilde{\theta}_1 = \frac{2}{n}\sum_{i=1}^{n} Mx_i = \frac{2}{n}\cdot nM\xi = \theta \Rightarrow$ оценка несмещенная

$D\tilde{\theta}_1 = D(\frac{2}{n}\sum_{i=1}^{n} x_i) = \frac{4}{n^2}\sum_{i=1}^{n} Dx_i = \frac{4}{n^2}\cdot nD\xi = \frac{4}{n}\cdot\frac{\theta^2}{12} = \frac{\theta^2}{3n} \xrightarrow[n\to\infty]{} 0$

$\Rightarrow$ оценка состоятельна

2) $\tilde{\theta}_2 = x_{min}$

$\forall\theta>0$ $M\tilde{\theta}_2 = Mx_{min} = \xi_{min} \sim \underbrace{1-(1-F(x))^n}_{\varphi(x)}$

$\varphi(x) = \varphi'(x) = n(1-F(x))^{n-1}F'(x) = n(1-\frac{x}{\theta})^{n-1}\cdot\frac{1}{\theta}\{[0,\theta]\}$

$Mx_{min} = \int_0^\theta x n \underbrace{(1-\frac{x}{\theta})}_{t}^{n-1}\frac{1}{\theta}dx = -\int_1^0 n(1-t)t^{n-1}\theta\, dt =$

$= n\theta(\int_0^1 t^{n-1}dt - \int_0^1 t^n dt) = n\theta(\frac{1}{n}-\frac{1}{n+1}) = \frac{\theta}{n+1} = M\tilde{\theta}_2 \neq \theta \Rightarrow$

$\Rightarrow \tilde{\theta}_2$ не явл. несмещенной

$\exists\ \tilde\theta_2' = (n+1)x_{min}$ – исправленная оценка. Тогда

$M\tilde\theta_2' = (n+1)Mx_{min} = \theta_2 \Rightarrow \tilde\theta_2'$ несмещенная оценка

$D\tilde\theta_2' = D((n+1)x_{min}) = (n+1)^2 D x_{min}$

$Mx_{min}^2 = \int_0^\theta x^2 n(1-\frac{x}{\theta})^{n-1}\frac{1}{\theta}dx = \int_0^1 n(1-t)^2 t^{n-1}\theta^2 dt =$

$= n\theta^2(\int_0^1 t^{n-1}dt - 2\int_0^1 t^n dt + \int_0^1 t^{n+1}dt) =$

$= n\theta^2(\frac{1}{n} - \frac{2}{n+1} + \frac{1}{n+2}) = n\theta^2 \cdot \frac{(n+1)(n+2) - 2n(n+2) + n(n+1)}{n(n+1)(n+2)} =$

$= \theta^2 \cdot \frac{n^2+3n+2-2n^2-4n+n^2+n}{(n+1)(n+2)} = \frac{2\theta^2}{(n+1)(n+2)}$

$D\tilde\theta_2 = \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} = \frac{n\theta^2}{(n+1)^2(n+2)}$

$D\tilde\theta_2' = \frac{n\theta^2}{n+2} \xrightarrow[n\to\infty]{} 0$

Проверим по определению: $P(|\tilde\theta_2' - \theta| \geq \varepsilon) \geq P(\tilde\theta_2' \geq \theta+\varepsilon) =$

$= P((n+1)x_{min} \geq \theta+\varepsilon) = P(x_{min} \geq \frac{\theta+\varepsilon}{n+1}) = 1 - \Phi(\frac{\theta+\varepsilon}{n+1}) =$

$= 1-(1-(1-\frac{\theta+\varepsilon}{\theta(n+1)})^n) = (1-\frac{\theta+\varepsilon}{\theta(n+1)})^n \xrightarrow[n\to\infty]{} e^{-\theta}\cdot e^{\theta+\varepsilon} \geq \varepsilon$

$\Rightarrow P(|\tilde\theta_2' - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0 \Rightarrow$ оценка $\tilde\theta_2'$ не состоятельна

Проверим $\tilde\theta_2$: $P(|x_{min} - \theta| \geq \varepsilon) = P(x_{min} \leq \theta - \varepsilon) = \Phi(\theta-\varepsilon) =$

$= -(1-F(\theta-\varepsilon))^n = 1-(1-\frac{\theta-\varepsilon}{\theta})^n = 1-(\frac{\varepsilon}{\theta})^n \xrightarrow[n\to\infty]{} 1$ при $\frac{\varepsilon}{\theta} < 1$

$\Rightarrow \tilde\theta_2 = x_{min}$ не явл. состоятельной

3) $\tilde\theta_3 = x_{max}$; $M\tilde\theta_3 = Mx_{max}$, $x_{max} \sim (F(x))^n = \Psi(x)$

$\psi(x) = \Psi'(x) = n(\frac{x}{\theta})^{n-1}\frac{1}{\theta}\{[0,\theta]\}$

$M x_{max} = \int_0^\theta x n \left(\frac{x}{\theta}\right)^{n-1} \frac{1}{\theta} dx = \frac{n}{\theta^n} \int_0^\theta x^n = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\theta \neq \theta$

$\Rightarrow \tilde{\theta}_3 = x_{max}$ не явл. несмещенной

$] \tilde{\theta}_3' = \frac{n+1}{n} x_{max}$ - исправленная оценка

$\Rightarrow M\tilde{\theta}_3' = \theta \Rightarrow \tilde{\theta}_3'$ - несмещенная оценка

$M x_{max}^2 = \int_0^\theta x^2 n \left(\frac{x}{\theta}\right)^{n-1} \frac{1}{\theta} dx = \frac{\theta^2 n}{n+2}$

$D x_{max} = \theta^2 \left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2}\right) = \frac{n\theta^2}{(n+2)(n+1)^2}$

$D\tilde{\theta}_3' = \frac{(n+1)^2}{n^2} \cdot \frac{n\theta^2}{(n+2)(n+1)^2} = \frac{\theta^2}{n(n+2)} \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_3'$ состоятельна

Проверим состоятельность по определению

$\tilde{\theta}_3$: $\forall \varepsilon > 0 \quad P(|\tilde{\theta}_3 - \theta| > \varepsilon) = P(\theta - x_{max} \overset{>}{\times} \varepsilon) = P(x_{max} < \theta - \varepsilon) =$

$= \left(\frac{\theta - \varepsilon}{\theta}\right)^n = \left(1 - \frac{\varepsilon}{\theta}\right)^n$

Т.к. $0 < 1 - \frac{\varepsilon}{\theta} < 1 \Rightarrow \left(1 - \frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_3 \xrightarrow{P} \theta$

$\Rightarrow \tilde{\theta}_3$ состоятельна

$\tilde{\theta}_3'$: Т.к. $x_{max} \xrightarrow{P} \theta$ и $\frac{n+1}{n} \xrightarrow[n\to\infty]{} 1$

Тогда по Т. о сходимости произведения:

$\tilde{\theta}_3' = \frac{n+1}{n} x_{max} = x_{max} + \frac{1}{n} x_{max}$

$\forall \varepsilon > 0 \quad P(|\tilde{\theta}_3' - \theta| > \varepsilon) \leq P(|x_{max} - \theta| > \frac{\varepsilon}{2}) + P(\frac{1}{n} x_{max} > \frac{\varepsilon}{2})$

$\xrightarrow{P} 0$ т.к. x_{max} состоятельна

$\frac{1}{n} x_{max} \leq \frac{\theta}{n} \Rightarrow$ для $n > \frac{2\theta}{\varepsilon} \hookrightarrow \frac{1}{n} x_{max} \leq \frac{\theta}{n} < \frac{\varepsilon}{2}$

$\Rightarrow P$ равна $0 \Rightarrow \tilde{\theta}_3' \xrightarrow{P} \theta \Rightarrow \tilde{\theta}_3'$ состоятельна

4) $\tilde{\theta}_4 = x_1 + \frac{1}{n-1}\sum_{i=2}^{n} x_i$

$M\tilde{\theta}_4 = Mx_1 + \frac{1}{n-1}\sum_{i=2}^{n} Mx_i = M\xi + \frac{1}{n-1}\sum_{i=2}^{n} M\xi = \frac{\theta}{2} + \frac{\theta}{2} = \theta$

$\Rightarrow \tilde{\theta}_4$ несмещенная

$D\tilde{\theta}_4 = Dx_1 + \frac{1}{(n-1)^2}\sum_{i=2}^{n} Dx_i = \frac{\theta^2}{12} + \frac{1}{n-1}\cdot\frac{\theta^2}{12} \xrightarrow[n\to\infty]{} 0$

$\Rightarrow$ не выполняется достаточное условие

$x_1 + \frac{1}{n-1}\sum_{i=2}^{n} x_i \xrightarrow{P} \xi + \frac{\theta}{2} \Rightarrow \tilde{\theta}_4$ не явл. состоятельной

(по ЗБЭ Хинчина $\exists M\xi_n < +\infty \Rightarrow \frac{1}{n}\sum_{i=1}^{n}\xi_n \xrightarrow{P} M\xi$)

5) $\tilde{\theta}_1 = 2\bar{x} \Rightarrow$, $D\tilde{\theta}_1 = \frac{\theta^2}{3n}$

$\tilde{\theta}_2' = (n+1)x_{min}$, $D\tilde{\theta}_2' = \frac{n\theta^2}{n+2}$ - не убывает с ростом n

$\tilde{\theta}_3' = \frac{n+1}{n}x_{max}$, $D\tilde{\theta}_3' = \frac{\theta^2}{n(n+2)} \underset{n\to\infty}{\sim} \frac{\theta^2}{n^2}$

$\tilde{\theta}_4 = x_1 + \frac{1}{n-1}\sum_{i=2}^{n} x_i$; $D\tilde{\theta}_4 = \frac{n\theta^2}{12(n-1)} \xrightarrow[n\to\infty]{} \frac{\theta^2}{12}$

$D\tilde{\theta}_3'$ убывает как $\frac{1}{n^2}$, $D\tilde{\theta}_1$ - как $\frac{1}{n}$ $\Rightarrow$ при $n\to\infty$

$\Rightarrow$ наиболее эффективной оценкой является $\tilde{\theta}_3'$